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Teaching Interns

March 18, 2026
Date of Teaching

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Teaching Internship Mentor

Grade 5-P/ WTh 7:30 AM -9:30 AM
Class and Schedule

March , 2026
Date of Submission

March, 2026
Date of Approval

LESSON PLAN in MATH 5
(1st Revision)

Content Standard	*The learners should have knowledge and understanding that:* - surface area of solid figures.
Performance Standard	By the end of the quarter, learners are able to determine the surface area of solid figures.
Learning Competencies	*The learners shall be able to:* a. illustrate and find the surface area of solid figures; and b. solve problems involving the surface area of solid figures.

I. Intended Learning Outcomes (Objectives)

At the end of the lesson, the Grade 5 learners with at least 80% accuracy will be able to:

Knowledge	Explain the formulas and components used to determine the surface area of at least three types of solid figures.
Skills	Construct a solid figure using materials available at home and calculate its surface area accurately.
Attitudes/Values	Appreciate the importance of accurately calculating surface area in real-life applications and describe its relevance.

II. Learning Content

Certification Date: 24 January 2024
Recertification due date: 24 January 2027
For verification of the certificate please Access www.gcl-intl.com (Certification Check and type the registration number)
Registration No. 52Q18778
The WORLD UNIVERSITY RANKINGS for INNOVATION
Times Higher Education Impact Rankings

A. Topic (s)	**Lesson 4.5 - Surface Area**
B. Key Concepts	• **Surface Area (SA)** refers to the total area of all the exposed faces or surfaces of a 3D solid figure. It is the sum of the areas of every individual shape that makes up the "skin" or "net" of the object. • A **rectangular prism** has 6 faces, consisting of three pairs of congruent (identical) opposite faces: Top/Bottom, Front/Back, and Left/Right. **Formula**: SA = 2(l x w) + 2(w x h) + 2(l x h). **Variables**: l = length w = width h = height. • A **cube** is a special type of rectangular prism where all 6 faces are identical squares. Because every edge has the same length, the calculation is simplified. **Formula**: SA = $6e^2$ **Variable**: e = the length of one edge • A **square pyramid** consists of 5 faces: one square base and four congruent triangular lateral faces that meet at a top point (the apex). **Formula**: SA = b^2 + 4(bh/2) **Variables**: b = the length of the side of the square base h = the slant height (the height of the triangular faces)
C. References *(Print, electronic, etc.)*	• Math 5 - Learners Module •

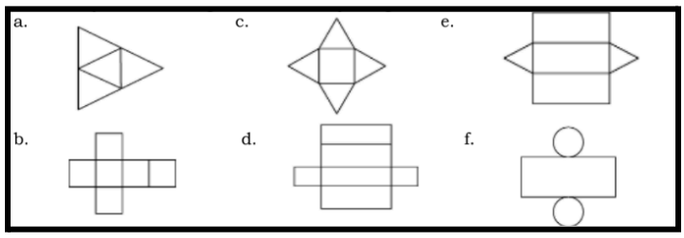
D. Instructional Materials	• Laptop • Canva Presentation • Google Docs • Google Forms
E. Integration *(Values, SDG/ESD, GAD, etc.)*	**Values:** Accuracy, patience, perseverance, responsibility. **SDG 4 (Quality Education):** Critical thinking, problem-solving, spatial reasoning. **SDG 9 (Industry, Innovation and Infrastructure):** Engineering and design skills. **SDG 11 (Sustainable Cities and Communities):** Resource planning, efficient material use.
F. Skills *(21st century skills)*	• Critical Thinking and Problem Solving • Communication Skills • Collaboration
III. Teaching and Learning Process (Experiences)	

Model: 4A Instructional Model
Total time allotment: 90 minutes

PRELIMINARIES (5 minutes)

The teacher begins the lesson with a prayer, followed by attendance checking. The teacher will express enthusiasm and encourage the students to stay curious, active, and engaged as they begin this new and dynamic part of their Math journey. After that, the teacher will ask some students to recall the topic they had last week.

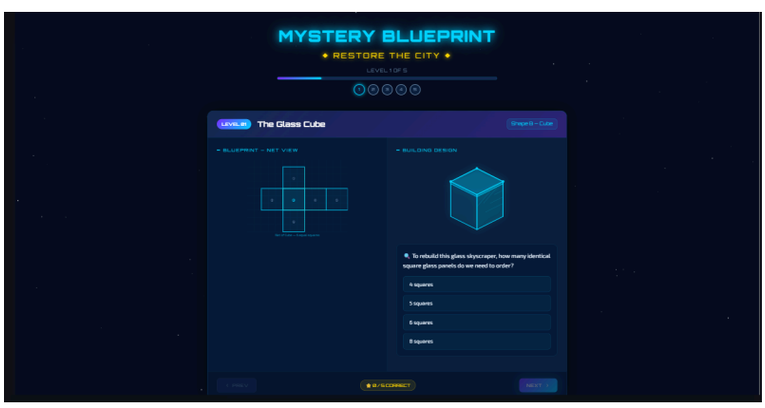
After the students share what they recall from the previous lesson, the teacher will affirm their responses by showing pictures of different figures. The teacher will not state the names of each 2D net, as these will be identified later during the lesson.



A. ACTIVITY

After the preliminaries, the teachers will shift to the next phase. This part will be an activity called **"Mystery Blueprint, Rebuild the City".**

Activity: "Mystery Blueprint, Rebuild the City"



The teacher will introduce the gamified scenario by telling students they are "Digital Architects" whose 3D city designs have been flattened into 2D blueprints (nets). After that, the teacher will display a 2D net on the screen (e.g., Net B, the Cube) and challenge the students to identify what solid figure it will form when folded.

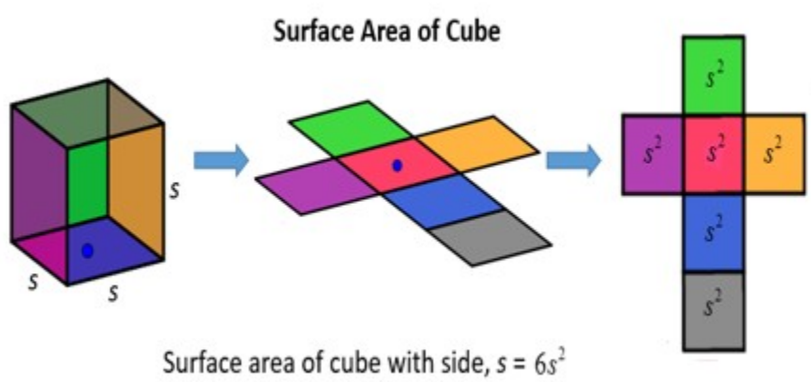
B. ANALYSIS

The teacher will then proceed to the analysis phase by presenting a **"Surface Area Mission"** question for each blueprint, such as: ***"To rebuild this glass skyscraper, how many identical square glass panels do we need to order?"*** to guide students in counting the faces of the net.

The analysis question will be asked after each shape is presented, and this process will be repeated for every blueprint until all shapes have been predicted, with students identifying each solid based on the number and arrangement of its faces. After all predictions have been completed, the teacher will conclude the game by explaining that the students have just determined the total "covering" of the shapes, which represents their surface area.

C. ABSTRACTION (20 minutes)

The teacher will guide the students how to derive the formulas for the surface area of different solids.



The teacher will give the surface area of the cube as an example, and explain the process first.

To get the surface area of a solid, one must first identify the surfaces in 2d. To do this, the teacher will flatten the 3d cube into a 2d projection so the surfaces are easier to identify.

After identifying the surfaces, the teacher will then ask the students to name the shape of each surface. In the cube's case, its surfaces are squares.

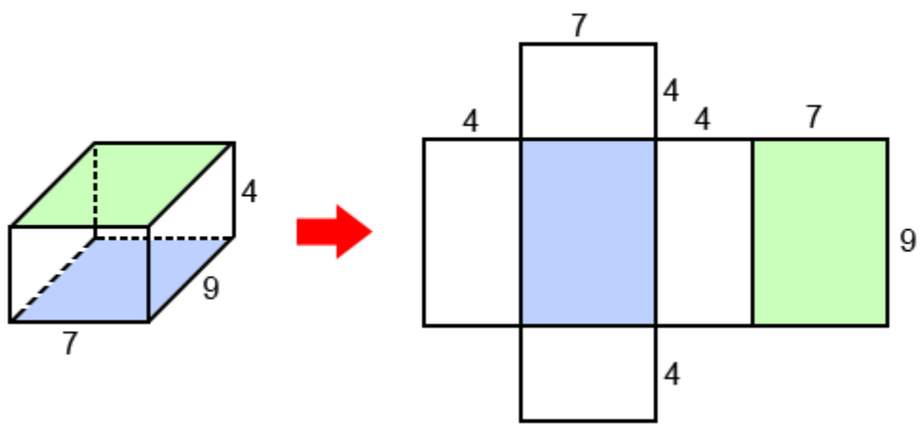
After identifying the shape of the surfaces, the teacher will then ask the students how to get the area of that shape, in the cube's case, the area of a

square is just the square of its sides.

After asking for the area, the teacher will ask the students to add all the areas of each surface together, in the cube's case, this will be $a^2 + a^2 + a^2 + a^2 + a^2 + a^2$ or $6a^2$.

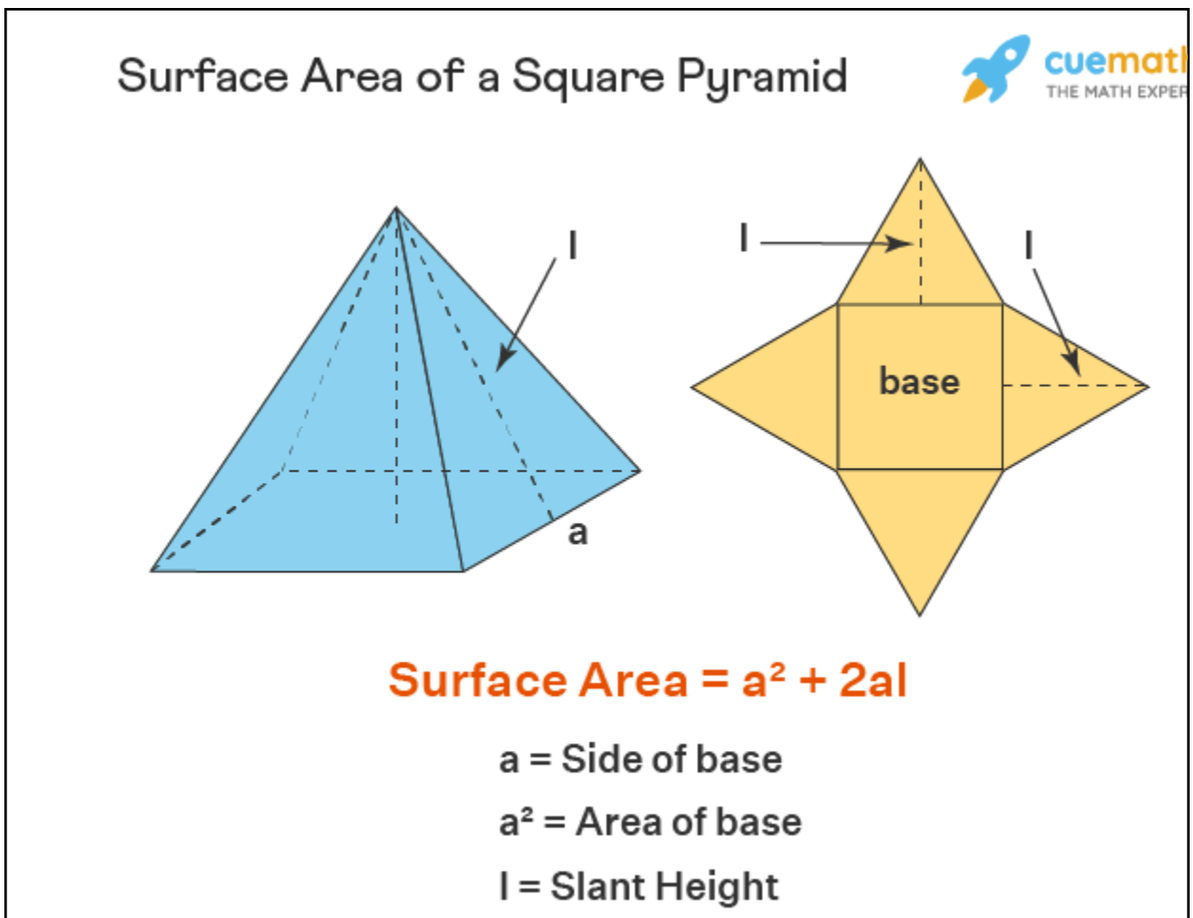
Therefore, the surface area of a cube is $6a^2$.

Next the teacher will show a picture of a rectangular prism



The teacher will have the students do the same steps to arrive to the following equation: SA = 2LW + 2 LH + 2 WH

The teacher will show a picture of a square pyramid.



The teacher will have the students use the same method to get a final equation of SA = a^2+2al.

D. APPLICATION (20 minutes)

The teacher will divide the class into 3 groups. The teacher will ask each group to solve for the surface area of the following figures by first deriving the surface area, and then using the derived formula to get the final answer.

Group 1: A cylinder with a radius of 5 cm, and a height of 15 cm.
Group 2: An equilateral triangle prism that has a height of 6 m.
Group 3: A triangular prism with a length of 7 cm and a height of 5 cm.

The group will present their solutions after 20 min of discussion or earlier.

Each group will also be challenged to derive, and explain the surface area of a sphere.

IV. Learning Assessment

Part 1: Multiple Choice Quiz

1. What is the surface area of a cube with an edge length of 3 cm?
   - **A) 54 cm²**
   - B) 27 cm²
   - C) 36 cm²
   - D) 18 cm²
2. A rectangular prism has a length of 5 m, width of 2 m, and height of 4 m. Calculate its surface area.
   - A) 40 m²
   - **B) 76 m²**
   - C) 52 m²
   - D) 88 m²
3. In the formula for a square pyramid, SA=b2+4(2bh), what does the variable **h** represent?
   - A) The vertical height from base to apex
   - B) The perimeter of the base
   - C) The area of the base
   - **D) The slant height of the triangular faces**
4. If a cube has a surface area of 150 in², what is the length of one edge (e)?
   - **A) 5 in**
   - B) 25 in
   - C) 6 in
   - D) 10 in
5. A square pyramid has a base side of 6 cm and a slant height of 10 cm. What is its total surface area?
   - A) 156 cm²
   - B) 120 cm²
   - **C) 156 cm²**
   - D) 276 cm²
6. How many pairs of congruent faces does a standard rectangular prism have?
   - A) 6
   - **B) 3**
   - C) 2
   - D) 1
7. Which formula correctly represents the surface area of a rectangular prism?
   - A) SA=l×w×h
   - B) SA=4(l+w+h)

- C) SA=6(l×w)
- **D) SA=2(lw)+2(wh)+2(lh)**

8. Calculate the surface area of a square pyramid with a base side of 4 ft and a slant height of 5 ft.
   - A) 40 ft²
   - B) 36 ft²
   - **C) 56 ft²**
   - D) 80 ft²
9. If you double the edge length of a cube from 1 unit to 2 units, what happens to the surface area?
   - **A) It increases from 6 to 24 (quadruples)**
   - B) It doubles
   - C) It stays the same
   - D) It triples
10. A rectangular box has dimensions 10x1x1. What is its surface area?
   - A) 10
   - B) 22
   - C) 40
   - **D) 42**

Part 2: Word Problems (5 Points Each)

1. Sarah is making a wooden jewelry box in the shape of a **rectangular prism**. The box is 10 inches long, 6 inches wide, and 4 inches tall. She wants to paint the entire outside of the box with gold leaf. If one small jar of gold leaf covers 50 square inches, how many jars will she need to buy to cover the entire surface?
2. An architect is designing a small decorative greenhouse in the shape of a **square pyramid**. The glass base is 8 feet wide on each side. The four triangular glass panels each have a slant height of 12 feet. What is the total square footage of glass needed to complete the entire structure (including the floor/base)?

V. Asynchronous Task (Enrichment or Agreement)

Activity: "Build & Measure Your Solid"

Objective: Construct a solid figure using materials at home and calculate its surface area.

Materials:

- Anything found at home (boxes, bottles, cups, papers, plastic containers, cardboard, etc.)
- Ruler or measuring tape
- Pencil and paper
- Tape, glue, or string (optional for assembling)

Instructions:

1. **Choose a Solid Figure:** Pick a shape (cube, rectangular prism, pyramid, cylinder, or cone) to represent with materials at home.
2. **Build Your Solid:** Using your chosen materials, construct the figure. Be creative—your materials don't have to be perfect as long as the shape is clear.
3. **Measure Dimensions:** Carefully measure the dimensions (length, width, height, radius) of your figure. Record them.
4. **Calculate Surface Area:** Using the appropriate formula for your solid figure, calculate its surface area. Show your work step by step.
5. **Reflect:** Take a photo of your constructed figure and your calculations. Write a short reflection on how this activity helped you understand surface area and its application in real life.

Rubric: Build & Measure Your Solid

Criteria	Excellent	Good	Satisfactory	Needs Improvement
Construction & Creativity (20 pts)	Solid figure is accurately constructed, clearly represents chosen shape, and shows high creativity using home materials	Solid figure is mostly accurate, represents chosen shape, and shows some creativity	Solid figure represents chosen shape but has minor inaccuracies; limited creativity	Solid figure is unclear or inaccurate; little to no creativity
Measurement & Calculation (15 pts)	All dimensions measured correctly; surface area calculated accurately with all steps shown	Most dimensions measured correctly; surface area mostly accurate with minor errors	Some dimensions measured correctly; surface area calculation partially correct	Dimensions measured incorrectly; surface area calculation mostly or completely wrong
Reflection (10 pts)	Thoughtful reflection clearly explains what was learned about surface area and real-life application	Reflection explains learning with some connection to real-life application	Reflection shows minimal explanation or connection to learning	Reflection missing or unclear; no connection to learning

Bonus (5 pts): Submitted before the deadline.

Grading Scale (50 pts):

- 45–50: Excellent
- 35–44: Good
- 25–34: Satisfactory
- Below 25: Needs Improvement